Technical Appendix: Query Performance at 2B Scale (ECAI)

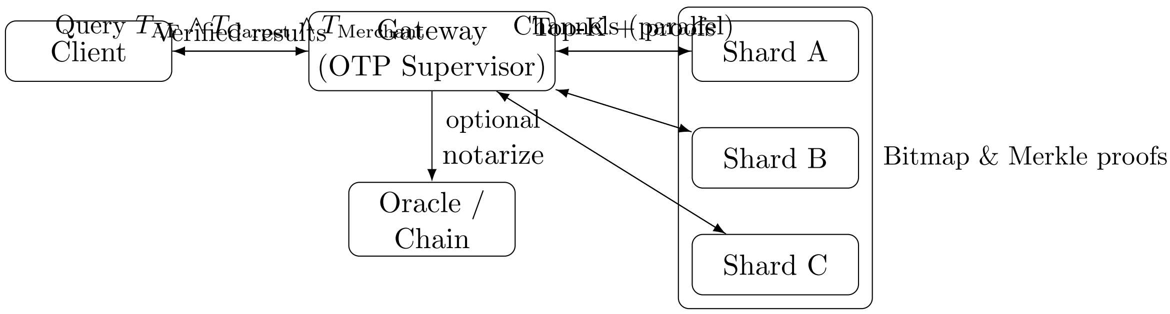
Goal. "Find a merchant in Afghanistan selling carpets" over a corpus of ∼2B facts, using **ECAI**'s SPOC→curve mapping, off-chain provable indexes, and **aeternity** channels for real-time, with optional on-chain notarization.

Assumptions (realistic production):

- Triples (*Subject, Predicate, Object, Context*) deterministically map to elliptic-curve points; canonical term IDs exist for `country:AF`, `sells:Carpet`, `type:Merchant`.
- Inverted indexes per term use compressed bitmaps (e.g., Roaring); postings Merklized per shard; global *index root* anchored to chain at regular intervals.
- Sharding by term-hash across 100–300 shards; queries fanned out over **aeternity state channels**; node stack in **Erlang/OTP**.

Query plan (AND of three terms):

1. **Normalize** to term IDs $T_{\mathrm{AF}}, T_{\mathrm{Carpet}}, T_{\mathrm{Merchant}}$.
2. **Route** to shards owning those terms via channels (parallel fan-out).
3. **Intersect** bitmaps per shard: $B(T_{\mathrm{AF}}) \cap B(T_{\mathrm{Carpet}}) \cap B(T_{\mathrm{Merchant}})$; return Top-K subjects + Merkle proofs.
4. **Verify** proofs against anchored index root; optionally **notarize** answer digest on-chain via oracle (TTL).



Latency budget (typical):

Stage	Warm/Local	Cold/WAN
Normalize terms & route (channels)	5–25 ms	20–80 ms
Shard bitmap intersections (parallel)	20–120 ms	80–400 ms
Fan-in, Top-K, proof packing	30–120 ms	80–300 ms
Client proof verification	5–30 ms	10–50 ms
Subtotal (realtime read)	**60–295 ms**	**190–830 ms**
Optional on-chain notarization (oracle TTL)	**3–10 minutes** (1–2 key blocks + 1 conf)	

Why it scales on 2B:

- High selectivity: three-term AND narrows hits dramatically.
- Bitmap math is word-level; intersections run per shard in parallel OTP processes.
- Returned proofs cover only touched postings; global integrity via anchored index root.

Operational levers:

- Tune shard count and postings compression; pre-compute frequent term intersections.
- Cache term bitmaps and Top-K heaps; keep hot shards in memory.
- Choose per-query: *channels for realtime* vs. *oracle TTL for notarized answers.*